

Application Build and Release Automation Lab Practice Exam 1

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Problem Scenario

GLA Campus Services is developing an Online Course Registration System. The development team uses Linux and Git to manage the project and collaborate on different application modules. The tasks below cover creating the project workspace, maintaining project files, using Git for version control, developing a feature using a separate branch, merging the feature, and resolving a Git merge conflict.

Question 1 — Linux Environment and File Management

5 Marks

Create a project directory:

- **<FirstName>-GLA-CourseRegistration**

Inside it create:

- src
- docs
- backup

Create a file: **course_info.txt**, containing:

- Student Name
- Roll Number
- Subject Name
- Project Name
- Semester

Perform suitable Linux commands to:

- Display the current working directory.
- Navigate into the project directory.
- Display the directory contents.
- Display the contents of course_info.txt.
- Display the project directory structure.

Answer

```
viyom@Viyom:/mnt/c/Users/viyom$ cd
viyom@Viyom:~$ pwd
/home/viyom
viyom@Viyom:~$ cd
viyom@Viyom:~$ pwd
/home/viyom
viyom@Viyom:~$ mkdir Viyom-GLA-CourseRegistration
viyom@Viyom:~$ ls
Viyom-GLA-CourseRegistration
viyom@Viyom:~$ cd Viyom-GLA-CourseRegistration
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ mkdir src docs backup
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ ls
backup docs src
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ touch course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ ls
backup course_info.txt docs src
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ nano course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ cat course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester
```

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ tree
.
├── backup
├── course_info.txt
├── docs
└── src

4 directories, 1 file
```

Question 2 — Git Repository and Version Control

6 Marks

Inside the project directory:

- Initialize a Git repository.
- Configure Git username and email if required.
- Check the repository status.
- Stage all project files.
- Create an initial commit.
- Modify course_info.txt by adding the course/program name and a project description.
- Check the repository status.
- View the changes made to the file.
- Stage and commit the modification.
- Display the commit history.

Answer

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/viyom/Viyom-GLA-CourseRegistration/.git/
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
```

```
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        course_info.txt

nothing added to commit but untracked files present (use "git add" to track)
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git add course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   course_info.txt
```

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git commit -m "Add Course Information"
[master (root-commit) e0b79bb] Add Course Information
1 file changed, 5 insertions(+)
create mode 100644 course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git log --oneline
e0b79bb (HEAD -> master) Add Course Information
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git show --stat
commit e0b79bb631fd5eb8ce8449b571fb9f652a860511 (HEAD -> master)
Author: Viyom Rana <viyomanuragrana@gmail.com>
Date:   Fri Sep 4 18:00:46 2026 +0000

    Add Course Information

    course_info.txt | 5 +++++
    1 file changed, 5 insertions(+)
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git diff
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ printf '\nCourse/Program Name: B.Tech Computer Science Engineering\nProject Description: Online Course R
egistration System for managing and searching courses\n' >> course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ cat course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
Project Description: Online Course Registration System for managing and searching courses
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git diff
```

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git diff
diff --git a/course_info.txt b/course_info.txt
index 5aaf50b..5b4c0de 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -3,3 +3,6 @@ Roll Number: 2415001904
 Subject Name: Application Build and Release Automation Lab
 Project Name: Online Course Registration System
 Semester: 5th Semester
+
+Course/Program Name: B.Tech Computer Science Engineering
+Project Description: Online Course Registration System for managing and searching courses
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git add course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   course_info.txt

viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git commit -m "Update Course Information"
[master 6e47b22] Update Course Information
 1 file changed, 3 insertions(+)
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git log --oneline
6e47b22 (HEAD -> master) Update Course Information
e0b79bb Add Course Information
```

Question 3 — Feature Branch Development

7 Marks

The development team wants to independently develop a Course Search Module.

Create a branch named: **course-search**

- Switch to the branch.
- Inside src, create: **course_search.txt**
- Add suitable information describing the Course Search Module.
- Stage and commit the file.
- Display all available branches.
- Display the current branch.
- Switch back to the main branch.
- Demonstrate that the feature was developed independently.

Answer

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git branch course-search
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git branch
course-search
* master
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git switch course-search
Switched to branch 'course-search'
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git branch
* course-search
master
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ cd src
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ touch course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ echo "Course Search Module"
Course Search Module
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ echo "Course Search Module" > course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ echo "Allows users to search and find available courses" >> course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ echo "Search can be performed using course name or course code" >> course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat course_search.txt
Course Search Module
Allows users to search and find available courses
Search can be performed using course name or course code
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git add course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git commit -m "Add Course Search Module"
[course-search cb72649] Add Course Search Module
1 file changed, 3 insertions(+)
create mode 100644 src/course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git status
On branch course-search
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch
course-search
* master
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git switch course-search
Switched to branch 'course-search'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch
* course-search
master
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch --show-current
course-search
```

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch
course-search
* master
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git ls-tree -r --name-only master
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git rev-parse --show-toplevel
/home/viyom/Viyom-GLA-CourseRegistration
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git log --oneline --all --decorate
cb72649 (course-search) Add Course Search Module
6e4f7b22 (HEAD -> master) Update Course Information
e0b79bb Add Course Information
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git show master:src/course_search.txt
fatal: path 'src/course_search.txt' does not exist in 'master'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git show course-search:src/course_search.txt
Course Search Module
Allows users to search and find available courses
Search can be performed using course name or course code
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ |
```

Question 4 — Merging and Conflict Resolution

8 Marks

Part A – Merge Feature

- Merge course-search into the main branch.
- Verify that course_search.txt is available.
- Display the Git history.

Answer

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git merge course-search
Updating 6e47b22..cb72649
Fast-forward
 src/course_search.txt | 3 +++
 1 file changed, 3 insertions(+)
 create mode 100644 src/course_search.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat course_search.txt
Course Search Module
Allows users to search and find available courses
Search can be performed using course name or course code
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git log --oneline --all --decorate
cb72649 (HEAD -> master, course-search) Add Course Search Module
6e47b22 Update Course Information
e0b79bb Add Course Information
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ |
```

Part B – Generate Conflict

Create a branch named: **course-update**

- Switch to course-update.
- Modify the same line of course_info.txt.
- Commit the modification.
- Switch back to the main branch.
- Modify the same line differently.
- Commit the main-branch modification.
- Attempt to merge course-update.
- Identify the merge conflict.
- Manually resolve the conflict.
- Remove all Git conflict markers.
- Stage the resolved file.
- Commit the resolved version.
- Verify the final repository status and history.

Answer

```

viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch course-update
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ okay next
okay: command not found
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git switch course-update
Switched to branch 'course-update'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ sed -i 's/^Project Description:./Project Description: Course registration system with course update
functionality./' ../course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat ../course_info.txt
bash: cat ../course_info.txt: No such file or directory
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat ../course_info.txt
bash: cat ../course_info.txt: No such file or directory
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat ../course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
Project Description: Course registration system with course update functionality.
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ okay done now next
okay: command not found
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git status
On branch course-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ../course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git add ../course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git commit -m "Update course information on course-update branch"
[course-update b98870f] Update course information on course-update branch
 1 file changed, 1 insertion(+), 1 deletion(-)
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git switch master
Switched to branch 'master'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git branch
  course-search
  course-update
* master
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ sed -i 's/^Project Description:./Project Description: Online Course Registration System with advanced course management./' ../course_info.txt

```

```

viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ sed -i 's/^Project Description:./Project Description: Online Course Registration System with advanced course management./' ../course_info.txt
sed: can't read ../course_info.txt: No such file or directory
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ ../course_info.txt
../course_info.txt: command not found
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ sed -i 's/^Project Description:./Project Description: Online Course Registration System with advanced course management./' ../course_info.txt
sed: -e expression #1, char 1: unknown command: '.'
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ sed -i 's|^Project Description:.|Project Description: Online Course Registration System with advanced course management.|' ../course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cat ../course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
Project Description: Online Course Registration System with advanced course management.
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ../course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git add ../course_info.txt
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git commit -m "Update course information on master branch"
[master a87ce60] Update course information on master branch
 1 file changed, 1 insertion(+), 1 deletion(-)
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.

```

```

GNU nano 8.7.1                               ../course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
<<<<<< HEAD
Project Description: Online Course Registration System with advanced course management.
=====
Project Description: Course registration system with course update functionality.
>>>>>> course-update

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark  M-J To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo     M-G Copy      ^B Where Was

```

```

viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   ../course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ cat ../course_info.txt
Student Name: Viyom Singh Rana
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
<<<<<< HEAD
Project Description: Online Course Registration System with advanced course management.
=====
Project Description: Course registration system with course update functionality.
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ cat ../course_info.txt
Student Name: Viyom Singh RanaRegistration/src$ nano ../course_info.txt
Roll Number: 2415001904
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester

Course/Program Name: B.Tech Computer Science Engineering
Project Description: Online Course Registration System with advanced course management.

viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git add ../course_info.txt
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git status
On branch master
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   ../course_info.txt

```

```

viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git commit -m "Resolve course informatiin merge conflict"
[master f102bc9] Resolve course informatiin merge conflict
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git status
On branch master
nothing to commit, working tree clean
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ git log --oneline --all --decorate
f102bc9 (HEAD -> master) Resolve course informatiin merge conflict
a87ce60 Update course information on master branch
b98870f (course-update) Update course information on course-update branch
cb72649 (course-search) Add Course Search Module
6e47b22 Update Course Information
e0b79bb Add Course Information
viiom@Viiom:~/Viiom-GLA-CourseRegistration/src$ |

```

Question 5 — Final Demonstration

4 Marks

Demonstrate:

- Complete Linux project structure.
- Git repository status and history.
- Feature branches.
- Successful feature merge.
- Conflict generation and resolution.

Answer

```
viyom@Viyom:~/Viyom-GLA-CourseRegistration/src$ cd ..
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ tree
.
├── backup
├── course_info.txt
├── docs
├── src
│   └── course_search.txt
└──

4 directories, 2 files
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git status
On branch master
nothing to commit, working tree clean
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git log --oneline --all --decorate
f102bc9 (HEAD -> master) Resolve course informatiin merge conflict
a87ce60 Update course information on master branch
b98870f (course-update) Update course information on course-update branch
cb72649 (course-search) Add Course Search Module
6e47b22 Update Course Information
e0b79bb Add Course Information
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ git branch
  course-search
  course-update
* master
viyom@Viyom:~/Viyom-GLA-CourseRegistration$ |
```